

UPI Fraud Analyzer

Data analytics report

Dataset	PaySim (from kaggle)
Fraud rate	147 fraud transaction(0.07%)
Model used	XGBoost, AUC - ROC
Tools used	Python, SQL, XGBoost, SHAP, Power BI

Summary:

India has over 117-118 billion transaction. As transaction number is growing, fraudulent cases are also increasing. This project is build with machine learning powered to detect fraud through raw data, by manipulating the data to identify fraud through different mechanism. Using SQL, Python and Power BI dashboard to analyze the huge data with relevant visualization.

1. Key outcome:

ROC AUC (Receiver Operating Characteristic Area Under the Curve) is 0.9962 and captures 93.9% of all fraudulent transaction within high risk tier meaning – model captures 138 of 147 as actual frauds.

2. Problem Statement

UPI fraud takes several forms – phishing links, account takeover, multiple small transaction to drain the account rapidly. Challenge the fraudulent transaction are rare (roughly 1 in 1354) and look similar to the normal transaction.

What are the gaps in current approach

- Most model flag the transaction that is above fixed amount (e.g. ₹ 50,000), but fraud cases are their for smaller amount as well.
- Since the cases are very rare identifying the fraud case from huge dataset can be inaccurate.

How this project addresses the gap

- Identifying by behavioural specific to UPI transaction patterns
- Using AUC – ROC as primary metrics for accuracy
- Using other identifying pattern such as night time transaction, transaction velocity, balance amount anomaly, new or unique senders, large amount etc.

This project also uses Power BI dashboard for insights.

3. Process:

3.1. Dataset

The dataset used for this project is from Kaggle PaySim money dataset, that consists of filtered 2,00,000 rows.

PaySim mirrors the real mobile payment behaviour and include labelled fraud cases. Converting the column names to UPI equivalent terminologies.

3.2. Feature Engineering

Several feature across 6 behavioural category:

Category	Features	Signals
Transaction velocity	txn_count_last_15min, is_velocity_burst	No. of transaction occurring through multiple accounts
Time-based risk	is_night_txn, time_risk_score	Most fraud occur during 1am – 4 am window period
Recipient signals	is_new_recipient, new_recipient_large_amount	New sender vpa transfer id high risk
Amount anomaly	amount_zscore, is_round_amount	Unusual amount being exchanged
Balance inconsistency	has_balance_error, balance_drop_ratio	The debited and credited amount don't match

Device & identity	is_new_device, txn_type_shift_flag	Sudden changes in device or signal account being taken over
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4. Results

Three models were trained, evaluated and compared. SMOTE (Synthetic Minority Over-sampling Techniques) is a method to balance datasets by creating new, fake data for the minority class, rather than just copying existing samples.

Model	AUC-ROC	Recall	Precision	F1-Score
Logic Regression(baseline)	0.9962	0.9310	0.0758	0.1403
Random forest	0.9778	0.9310	0.9643	0.9474
XGBoost(More accurate)	0.9971	0.9310	0.9643	0.9474

XGBoost was the best model out of three. The scale_pos_weight parameter was set to the ratio of non fraud to fraud cases (1354:1), allowing the model to narrow down the fraud rate with precision.

5. Insights

5.1 Balance Inconsistency:

The most predictive feature was has_balance_error, cases where post transaction balance did not match with the expected value after deducting the transfer amount.

This feature had a SHAP importance 0.32 and was present in over 95% of confirmed cases.

5.2 Fraud spikes at night:

Transactions initiated between 1am and 4am showed a fraud rate of 3.8% — nearly 4 times the baseline rate of 1.03%.

5.3 Transaction velocity:

Accounts that sent 3 or more transactions within a 15-minute window — flagged by is_velocity_burst — had a fraud rate of 0.79%.

5.4 New recipient and large amount:

The combination feature new_recipient_large_amount — where a sender transacts with a never-before-seen VPA for an amount above the 75th percentile — had a precision of 79% for fraud.

5.5 CASHOUT and P2P :

CASHOUT and P2P showed 0.43% and 0.31% respectively. This reflects the fact that fraudsters prefer to move money to third-party accounts rather than make traceable merchant purchases.

6. Risk Tier Analysis

Tier	Transactions	Fraud captured	Recommended action
High	0.1% of total	98.6% of all fraud	Block & review
Medium	0.002% of total	7.1% of all fraud	Flag & monitor
Low	99% of total	1.6% of all fraud	Allow

7. Power BI dashboard

- The interactive dashboard deliver five key views:
- KPI rows: Fraud rate%, fraud capture %, fraud count, high risk count, model AUC-ROC
- Fraud rate by hour 1am–4am window highlighted in coral
- Risk tier donut chart — Low / Medium / High split across all transactions
- Top flagged transactions table — sortable by risk score with drill-through
- SHAP signal importance bar — shows which features drive fraud predictions

8. Resources and libraries used

Component	Tools & libraries
Data wrangling	Python 3.11, pandas, numpy, SQLAlchemy, PyMySQL
Feature engineering	pandas, scipy, rolling window functions
Modelling	scikit-learn, XGBoost, imbalanced-learn (SMOTE)
Explainability	SHAP (TreeExplainer, waterfall plots, beeswarm plots)
Visualisation	matplotlib, seaborn, Power BI Desktop
Database	MySQL 8.0 (upi_fraud_db schema)
Version control	Git, GitHub (public portfolio repository)